

DSD Axioms, Definitions, Lemmas, Theorems, and Open Proof Obligations

Purpose

This document strips the structural theory down to a formal dependency ledger. The goal is to expose hidden assumptions, circularity, undefined terms, and points where known mathematics is being invoked. Items are labeled AXIOM, DEFINITION, DERIVED LEMMA, KNOWN THEOREM INPUT, CONDITIONAL THEOREM, NO-GO, or OPEN PROOF OBLIGATION.

Level 0 — Primitive Distinction Structure

DEFINITION D0.1 — Orthogonality space

Let X be a set equipped with a symmetric distinguishability/orthogonality relation $\perp$.

AXIOM A0.1 — Irredundancy

$$x \perp y \Rightarrow x = y.$$

No two distinct states possess the same complete distinction profile.

AXIOM A0.2 — Distinction Self-Duality

For every admissible physical map $f:A \rightarrow B$ and every $b \in B$, there exists a unique $f^\dagger(b) \in A$ such that

$$f(a) \perp b \Leftrightarrow a \perp f^\dagger(b)$$

for all $a \in A$.

DERIVED LEMMA L0.1 — Dagger identity

$$I^\dagger = I.$$

DERIVED LEMMA L0.2 — Contravariant composition

$$(g \circ f)^\dagger = f^\dagger \circ g^\dagger.$$

DERIVED LEMMA L0.3 — Involution

$$(f^\dagger)^\dagger = f,$$

provided the map class and irredundancy assumptions are closed under the required profiles.

OPEN PROOF OBLIGATION O0.1

State the admissible map category exactly and prove all three dagger laws without importing Hilbert-space adjoints in the definitions.

Level 1 — Quantum Representation

DEFINITION D1.1 — Orthogonal closure

For $S \subseteq X$ define

$S^\perp = \{y : y \perp x \text{ for all } x \in S\},$

$K(S) = S^\perp^\perp.$

KNOWN THEOREM INPUT K1.1

Under appropriate point-closedness, irreducibility, rank, orthomodularity, and representation hypotheses, orthogonality spaces can be represented by projective Hermitian geometries.

CONDITIONAL THEOREM T1.1 — DSD-to-Hilbert representation

If the DSD orthogonality space satisfies the exact hypotheses of the selected representation theorem, then the completed proposition structure is represented in a Hilbert-space-like geometry.

OPEN PROOF OBLIGATION O1.1

Minimize the representation hypotheses and identify which are consequences of DSD versus additional physical assumptions.

KNOWN THEOREM INPUT K1.2 — Probability representation

For Hilbert spaces satisfying the relevant dimensional and regularity conditions, Gleason-type theorems yield

$\mu_p(P) = \text{Tr}(\rho P).$

KNOWN THEOREM INPUT K1.3 — Complex composition

Under the adopted local-tomographic/compositional assumptions, ordinary complex tensor composition

$H_{AB} = H_A \otimes H_B$

is selected over real and quaternionic alternatives.

OPEN PROOF OBLIGATION O1.2

Show whether the composition assumptions can be motivated independently from self-contained distinction rather than adopted solely because they select complex quantum theory.

Level 2 — Minimal Complex Binary Geometry

DEFINITION D2.1

Take the smallest nontrivial complex system $H_2 = \mathbb{C}^2$.

DEFINITION D2.2

$V = \text{Herm}_2(\mathbb{C})$.

DERIVED LEMMA L2.1 — Complement

$J(X) = \text{Tr}(X)I - X$,
with $J^2 = I$.

DERIVED LEMMA L2.2 — Lorentz bilinear

Define

$g(X, Y) = 1/2 \text{Tr}[XJ(Y)]$.

Writing $X = x^0 I + x^i \sigma_i$ gives

$g(X, X) = (x^0)^2 - |x|^2 = \det X$.

KNOWN THEOREM INPUT K2.1

$\text{Herm}_2^+(\mathbb{C})$ is isomorphic to the four-dimensional future Lorentz cone, and
 $\text{SL}(2, \mathbb{C})/\{\pm I\} \cong \text{SO}^+(1, 3)$.

CONDITIONAL RESULT T2.1

If the minimal physically realized distinction carrier is a complex qubit, its real self-adjoint carrier has local 3+1 Lorentz geometry.

CAUTION C2.1

This does not by itself derive macroscopic spacetime dimension. The four-dimensional realization argument must remain logically independent.

Level 3 — Primitive Distinction Dynamics

DEFINITION D3.1 — Complementary covariant source law

$E(dQ) = 2(I - Q)dvFS(Q)$.

DEFINITION D3.2

$WP(dQ) = 2\text{Tr}[P(I - Q)]dvFS(Q)$.

DERIVED LEMMA L3.1 — Bloch form

$w(n, m) = 1 - n \cdot m$.

DERIVED LEMMA L3.2 — Spectrum

The associated Markov operator has $\text{spec}(T)=\{1,-1/3,0\}$.

DERIVED COROLLARY L3.3

The isotropic profile is stationary; the nontrivial $l=1$ orientation mode contracts by magnitude $1/3$ per step.

NO-GO N3.1

The quantity $3^{\Delta-d}$ must not be used as a one-step record-space jump law. On natural Hamming record spaces, $P_{xy} \propto 3^{\Delta-dR}$ produces extensive one-step changes and is nonlocal.

OPEN PROOF OBLIGATION O3.1

Prove the uniqueness of the adopted complementary covariant POVM from the stated DSD physical principles without hidden normalization assumptions.

Level 4 — Thermodynamic Realization

DEFINITION D4.1 — Redundant realization

A distinction is thermodynamically realized when many independent internal fragments carry recoverable records of the same orthogonal alternative with error tending to zero in the appropriate limit.

MODEL D4.2 — Record isometry

$VP(R)=P \otimes |1\rangle^{\Delta} \otimes R + (I-P) \otimes |0\rangle^{\Delta} \otimes R$.

DERIVED/KNOWN RESULT L4.1

For fragment fidelity $q < 1$, record ambiguity can decay exponentially, schematically $FR \leq q^{\Delta R}$.

CONDITIONAL RESULT T4.1

In a thermodynamic/quasi-local limit, redundantly recorded classical alternatives may become operationally disjoint sectors under finite local operations.

DEFINITION D4.3 — Realized event record algebra

Let R_e denote the objective record algebra associated with event e .

DEFINITION D4.4 — Record causal precedence

$e \leq f$ iff $R_e \subseteq R_f$,
modulo equal-record equivalence.

OPEN PROOF OBLIGATION O4.1

Prove that this ordering agrees with operational causal influence in the intended realized sector.

REFINEMENT R4.1 — Spacelike independence

Record incomparability is necessary but insufficient. True independence additionally requires no signaling in either direction and RPI diamond closure.

Level 5 — Growth and Locality

NO-GO N5.1 — Bounded-parent theorem

If realized event n has k_n direct parents and $k_n \leq K$ uniformly, persistent growth gives $N_0(N) = N_0(\text{seed}) + \sum_n k_n = O(N)$.

This cannot match a four-dimensional interval phase requiring $N_0 \sim CN^{3/2}$.

COROLLARY N5.2

Binary DSD microinteraction cannot be identified one-to-one with thermodynamically realized causal events.

DEFINITION D5.1 — Record metric

$dR(x,y) = |R_x \Delta R_y|$.

DEFINITION D5.2 — Primitive local ancestry

$x \sim y$ iff $dR(x,y) = 1$,

or a finite-range weighted generalization.

DEFINITION D5.3 — Reversible local generator

$(Lf)(x) = \sum_{y \sim x} c_{xy}[f(y) - f(x)]$,

with $c_{xy} = c_{yx} \geq 0$.

KNOWN THEOREM INPUT K5.1

Volume doubling plus a scale-two Poincare inequality imply Gaussian heat-kernel behavior for suitable reversible graph walks.

OPEN PROOF O5.1

Derive or justify the required regularity conditions from the vacuum record geometry instead of assuming diffusion by fiat.

Level 6 — Operational Vacuum

DEFINITION D6.1 — Operational vacuum

For all x, y and every finite radius R ,
 $B^*R(x) \cong B^*R(y)$,
including all locally observable record, DSD, and transition decoration.

CONDITIONAL LEMMA L6.1

In a connected locally finite setting, compatible rooted-ball equivalences at every radius extend to global automorphisms, so operational vacuum implies homogeneity.

KNOWN THEOREM INPUT K6.1

Within the relevant median/cubical non-exponential sector, homogeneous geometries are coarsely lattice-like:

$G_{\text{vac}} \sim \mathbb{Z}^m \times F$
with finite F .

CAUTION C6.1

This does not exclude every mathematically possible exponential-growth median phase. Claims must remain scoped to the stated sector unless a stronger theorem is obtained.

Level 7 — Dimension Selection

DEFINITION D7.1 — Spectral dimension

Assume on-diagonal heat-kernel scaling
 $p_t(x, x) \sim t^{-(d_s/2)}$.

DEFINITION D7.2 — Intersection local time

$IL = \sum_{s, t \leq L} 1_{\{X_s = Y_t\}}$
for independent ancestry histories X, Y .

KNOWN SCALING INPUT K7.1

Under ordinary diffusion,
 $E[IL] \sim$
 $L^{(2-d_s/2)}$ for $d_s < 4$,
 $\log L$ for $d_s = 4$,
 $O(1)$ for $d_s > 4$.

OPEN DEFINITION O7.1 — Witness independence

Replace the current verbal condition with a precise asymptotic quantity, such as mutual information, covariance, channel distinguishability, or overlap between separated witness sigma-algebras.

OPEN DEFINITION O7.2 — Persistent corroborability

Replace the verbal condition with a precise probability or information-theoretic requirement tied to shared ancestry intersections.

FLAGSHIP CONJECTURE C7.1

The final non-circular definitions imply
witness independence $\Rightarrow ds \geq 4$,
persistent corroborability $\Rightarrow ds \leq 4$,
hence $ds = 4$.

OPEN PROOF O7.3

Prove C7.1 and classify counterexamples outside the regular diffusive basin.

Level 8 — Realization-Path Independence

AXIOM A8.1 — RPI

Different linear extensions/foliations of the same fully realized intrinsic causal history produce the same physical map modulo gauge acting trivially on fixed physical boundary observables.

DEFINITION D8.1 — Elementary RPI diamond

For incomparable realized events e, f above stem l :

$$[R_f(l+e) \circ R_e(l)]_{\text{phys}} = \\ [R_e(l+f) \circ R_f(l)]_{\text{phys}}.$$

KNOWN COMBINATORIAL INPUT K8.1

Any two linear extensions are connected by adjacent swaps of incomparable elements.

DERIVED/CONDITIONAL RESULT L8.1

For canonical tensor-factor swaps on independent complex quantum sectors, the symmetric-group relations close exactly.

DERIVED RESULT L8.2

Projective $U(1)$ phases cancel in adjoint action on density operators:
 $\text{Ad}U_g \circ \text{Ad}U_h = \text{Ad}U_{gh}$.

OPEN PROOF O8.1

Extend finite-sector anomaly freedom to thermodynamic, continuum, and boundary-sensitive sectors.

Level 9 — Hypersurface Reconstruction

CONDITION A9.1 — Smooth record-front scaling

A maximal realized antichain admits tangent displacements

$$Ea = \ell e_a + O(\ell^2)$$

with a smooth nondegenerate limit.

DEFINITION D9.1 — Spatial metric

$$q_{ab} = -g(e_a, e_b).$$

DEFINITION D9.2 — Future normal

$$g(n, n) = 1,$$

$$g(n, e_a) = 0.$$

DEFINITION D9.3 — Normal realization deformation

$$\delta N X = N n.$$

DERIVED LEMMA L9.1 — Normal tilt

Preserving normalization and orthogonality gives

$$\delta N n = q^{ab} \partial_b N e_a.$$

DERIVED LEMMA L9.2 — Normal-normal bracket

$$[\delta N, \delta M] X =$$

$$q^{ab} (M \partial_b N - N \partial_b M) e_a.$$

DERIVED LEMMA L9.3 — Full geometric deformation algebra

Together with tangential relabelings,

$$[\delta N_{\text{vec}}, \delta M_{\text{vec}}] = \delta [N_{\text{vec}}, M_{\text{vec}}],$$

$$[\delta N_{\text{vec}}, \delta M] = \delta (L_{N_{\text{vec}}} M),$$

and L9.2 provide the geometric hypersurface-deformation algebra.

OPEN PROOF O9.1

Make the discrete finite-difference convergence rigorous and determine whether any $O(1)$ anomaly or extra structure function survives as $\ell \rightarrow 0$.

Level 10 — Canonical GR Limit

CONDITION A10.1

The smooth metric-only geometric sector admits a regular semiclassical symplectic phase space after gauge quotient.

KNOWN THEOREM INPUT K10.1

Under Hojman-Kuchar-Teitelboim-type locality and geometrodynamical assumptions, a canonical representation of the hypersurface-deformation algebra yields ADM/Einstein dynamics up to constants G and Λ .

CONDITIONAL THEOREM T10.1

If A10.1 and K10.1 apply to the DSD/RPI continuum sector, the infrared gravitational action is Einstein-Hilbert plus higher-order corrections.

OPEN PROOF O10.1

Show that no additional gapless scalar, vector, torsion, or other geometric order parameter survives the microscopic-to-continuum limit.

Open Quantitative Frontier

Q1. Derive G .

Q2. Derive Λ .

Q3. Derive the absolute microscopic scale ℓ^* .

Q4. Calculate finite- ℓ and higher-derivative correction coefficients.

Q5. Derive the matter/gauge excitation spectrum.

Q6. Determine the basin of attraction of the four-dimensional vacuum.

Dependency Rule

No later calculation may repair a failure by importing a desired result from a downstream level.

In particular:

dimension four may not be assumed in the realization proof;

Einstein dynamics may not be used to justify RPI;

observed G or Λ may not be used to construct microscopic formulas;

continuum covariance may not be used to prove the discrete RPI law.

Priority Order

1. Formalize O7.1 and O7.2.

2. Prove or kill C7.1.

3. Complete O9.1.
4. Complete O8.1.
5. Only then begin quantitative microscopic calibration.